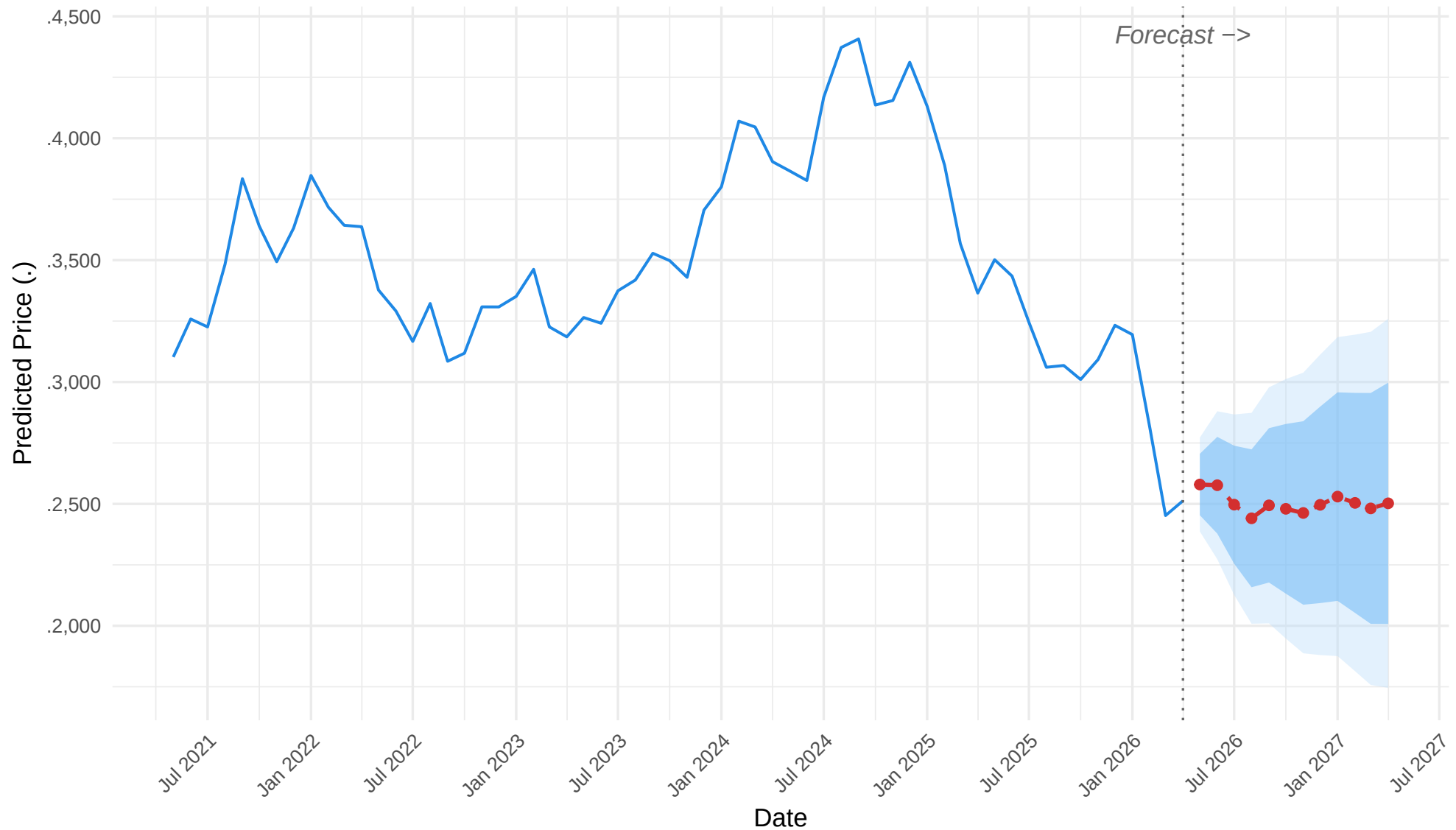


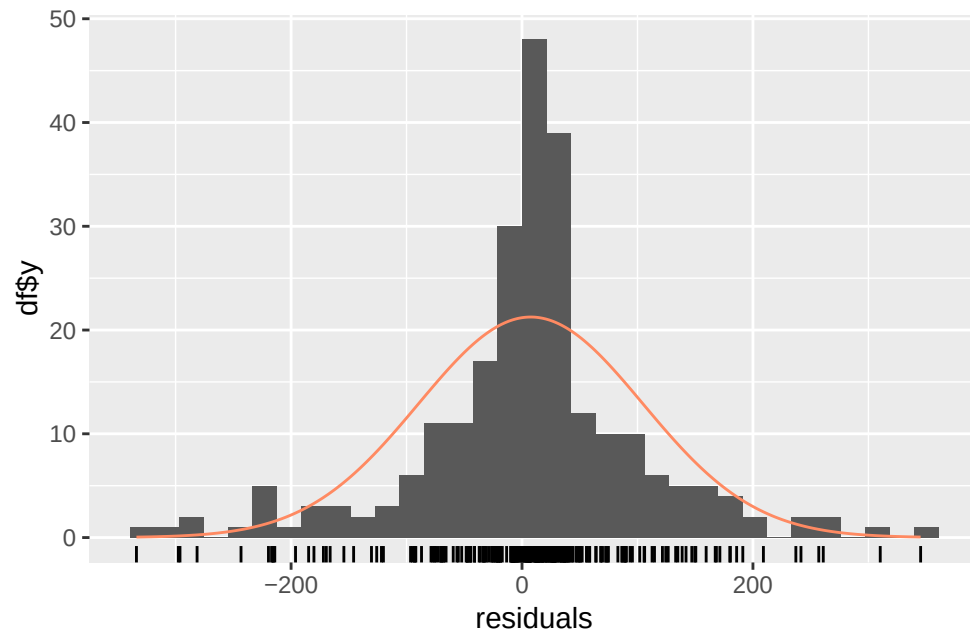
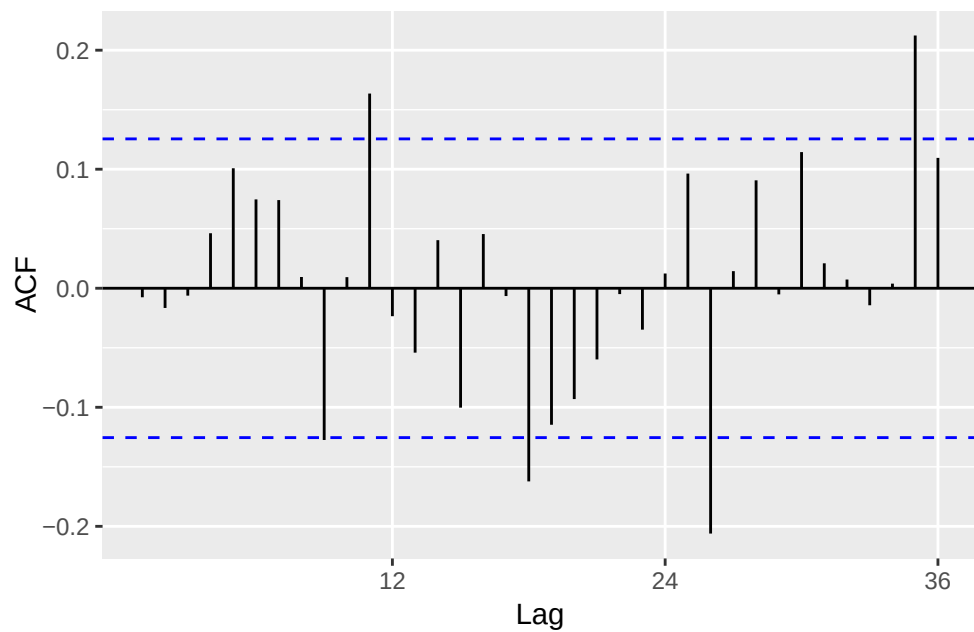
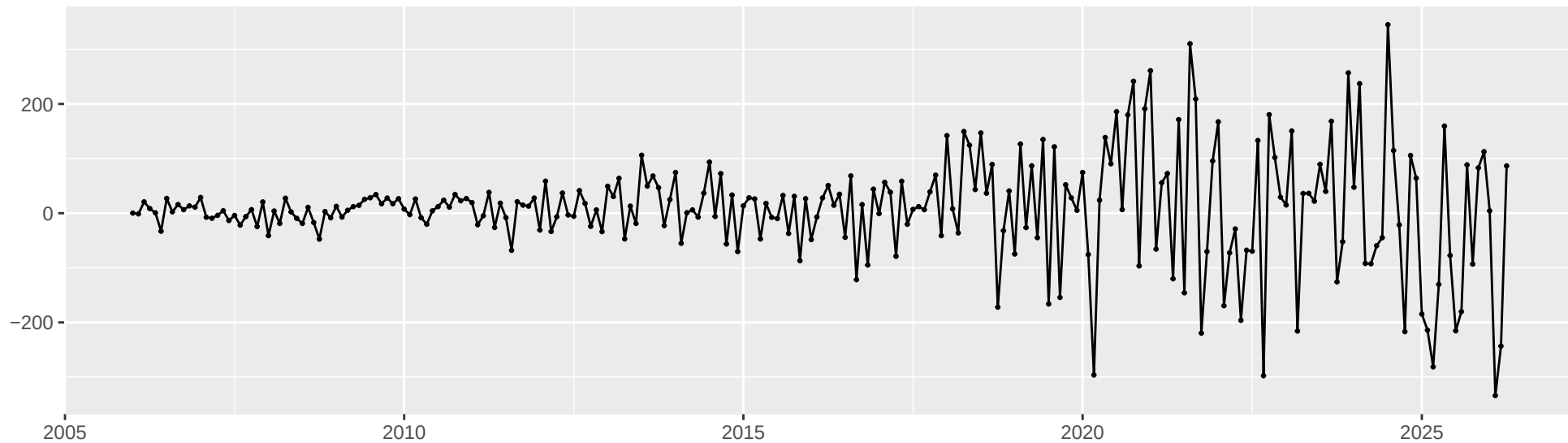
TCS – 12-Month ARIMA Forecast

Model: ARIMA(1,1,1)(1,0,1)[12]

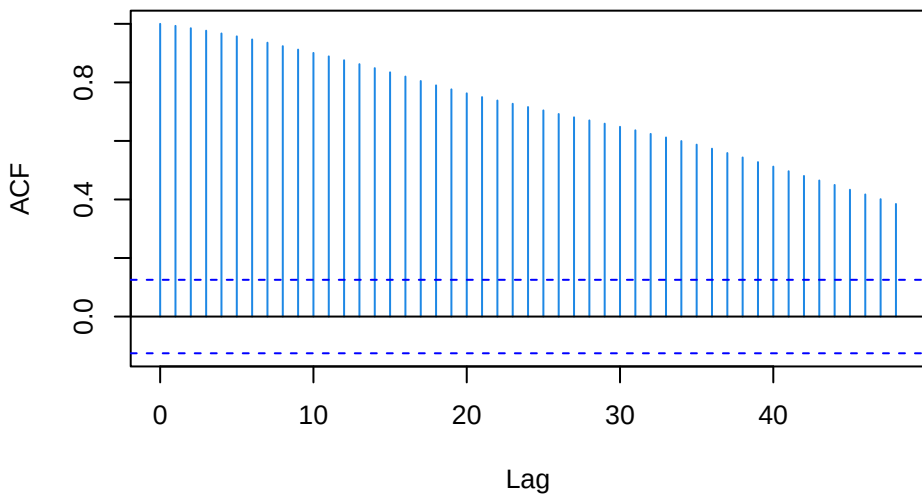


Shaded regions: 80% (dark) and 95% (light) confidence intervals

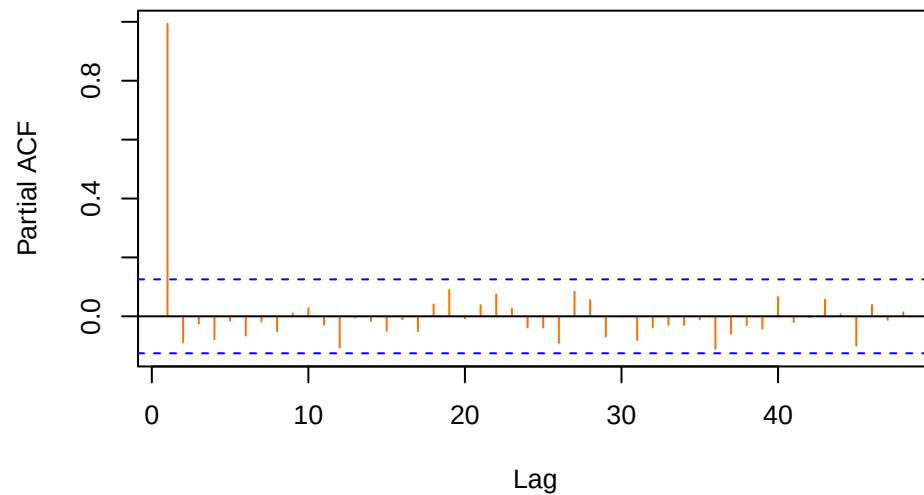
Residuals from ARIMA(1,1,1)(1,0,1)[12]



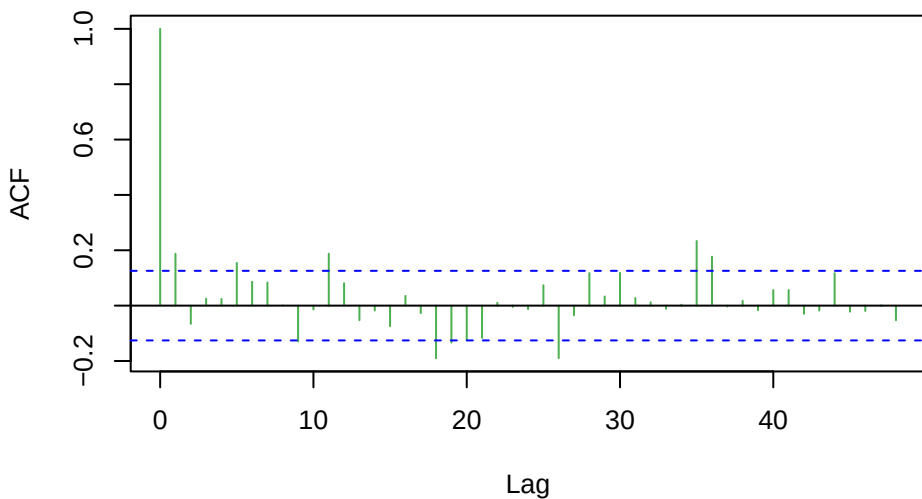
ACF – TCS Raw Monthly Prices



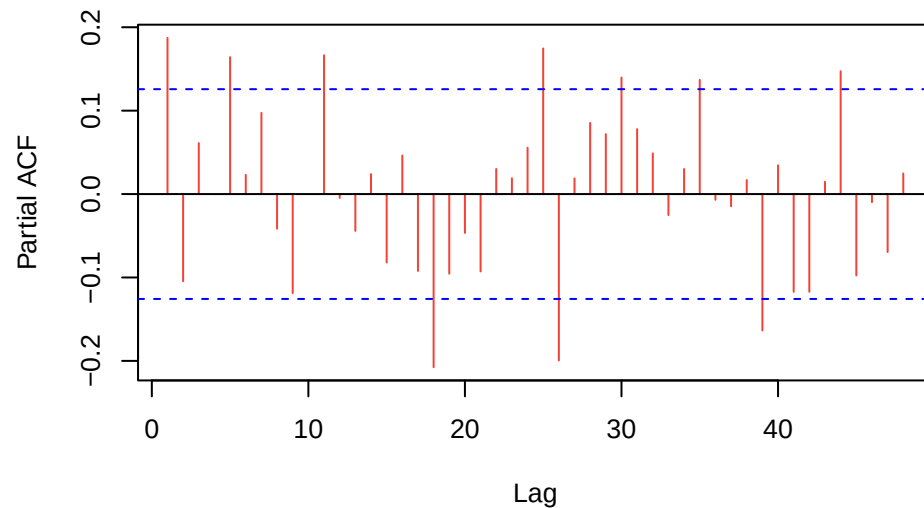
PACF – TCS Raw Monthly Prices



ACF – First Differenced



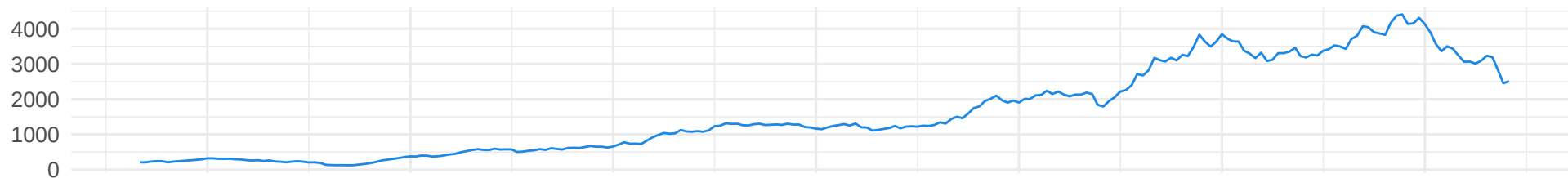
PACF – First Differenced



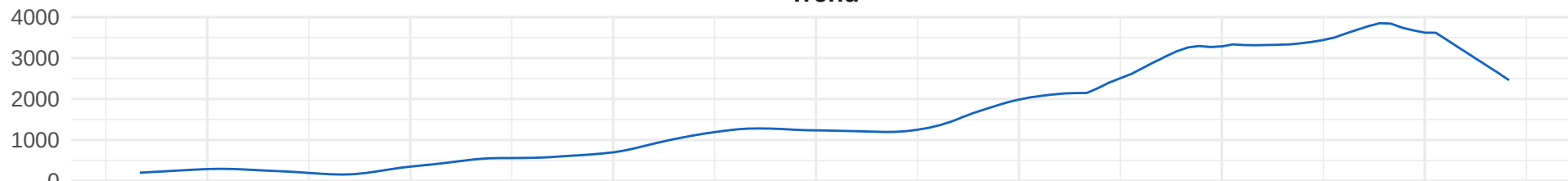
TCS – Complete STL Decomposition

Observed = Trend + Seasonal + Residual

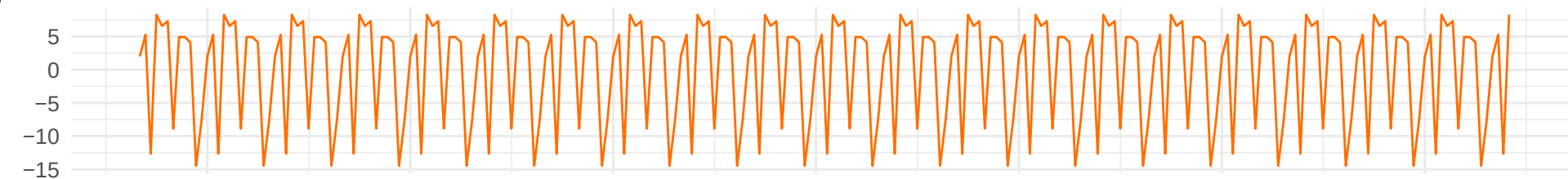
Observed



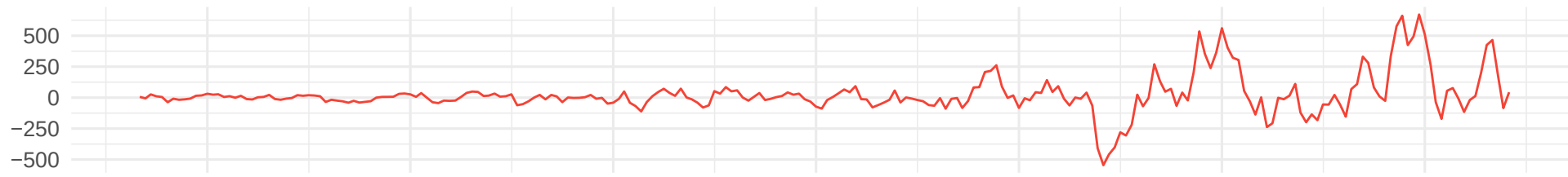
Trend



Seasonal



Residual



2007

2010

2013

2016

2019

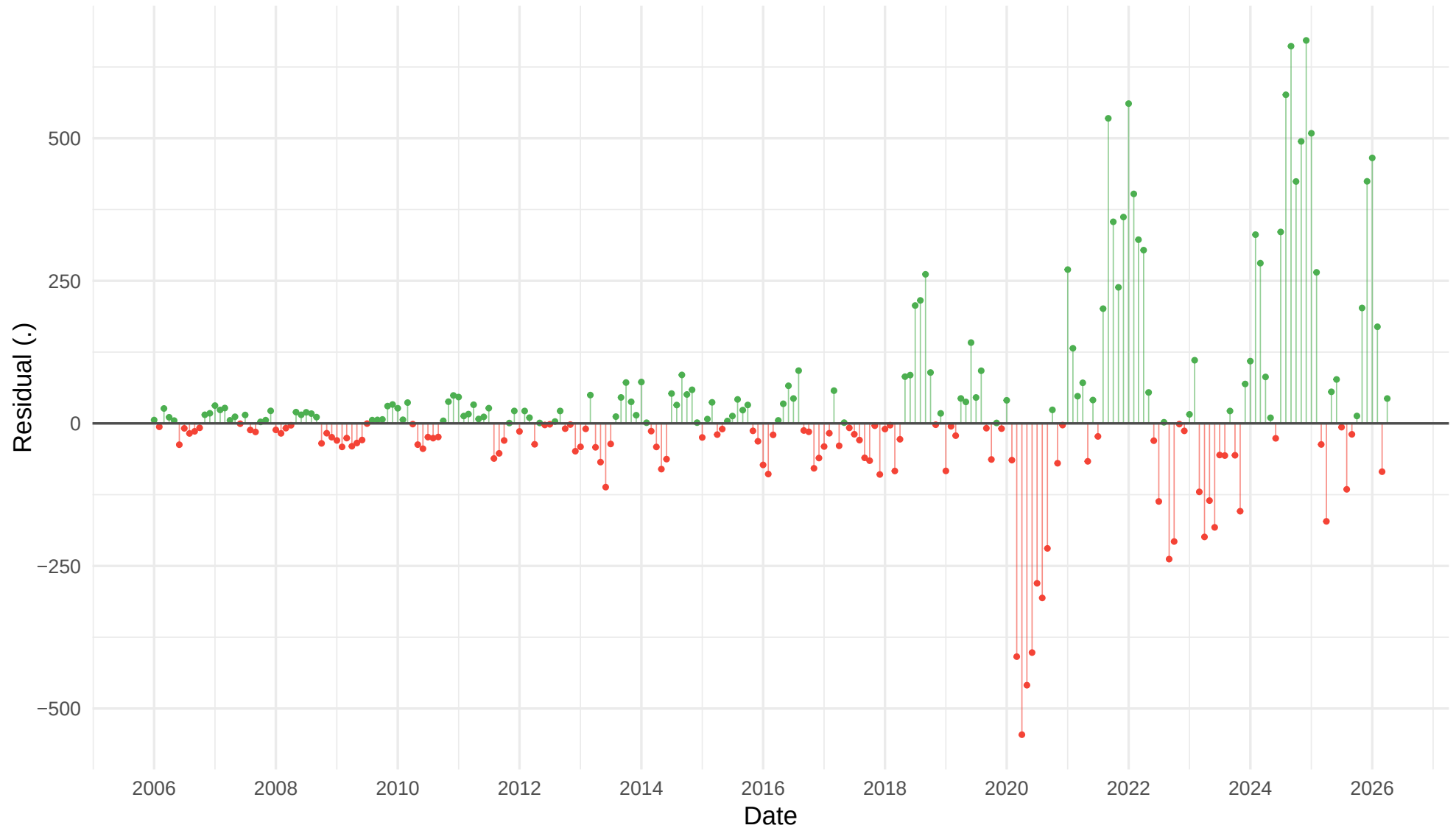
2022

2025

Date

TCS – Residual (Remainder) Component

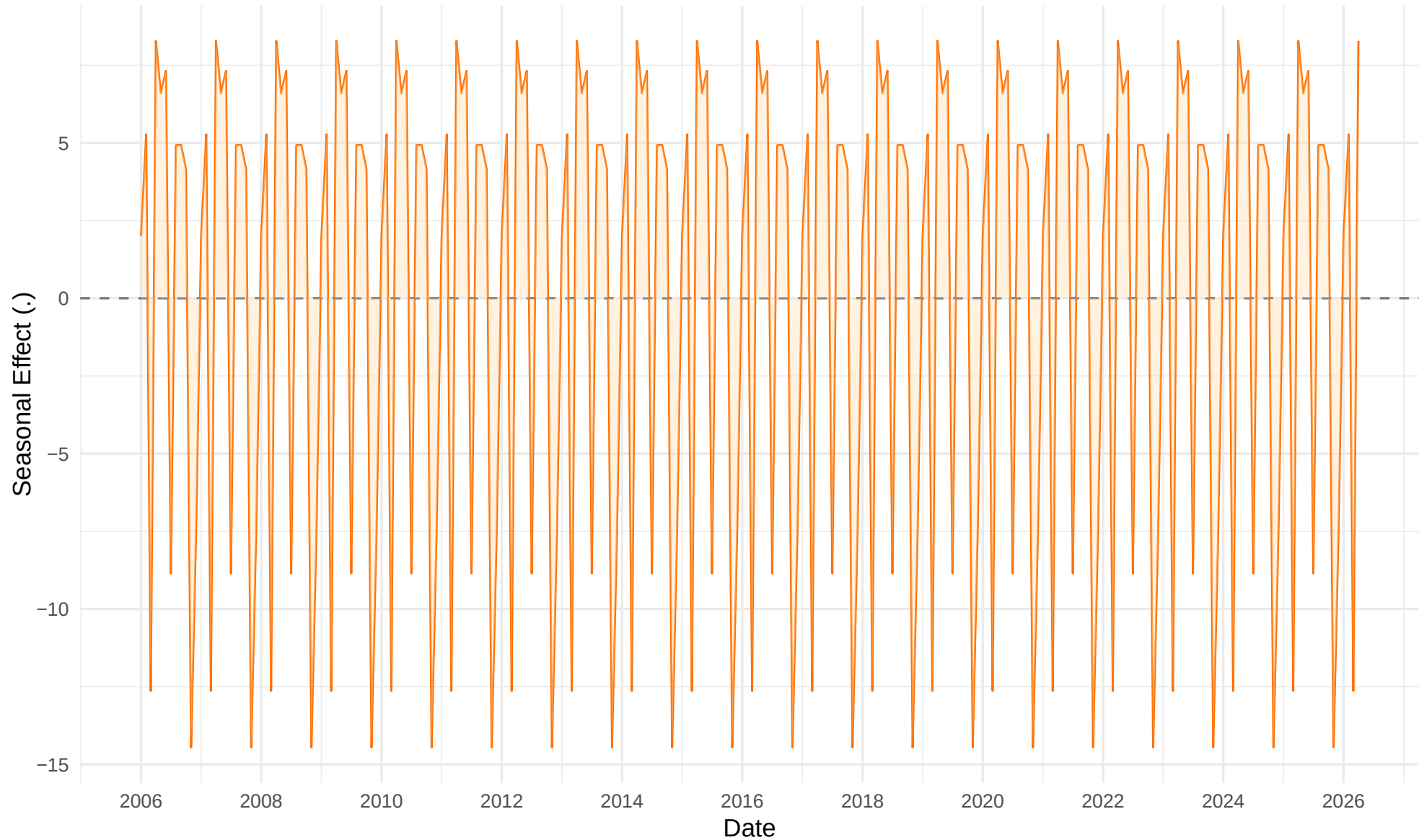
Random noise after removing trend and seasonality



Large residuals indicate unexpected price movements

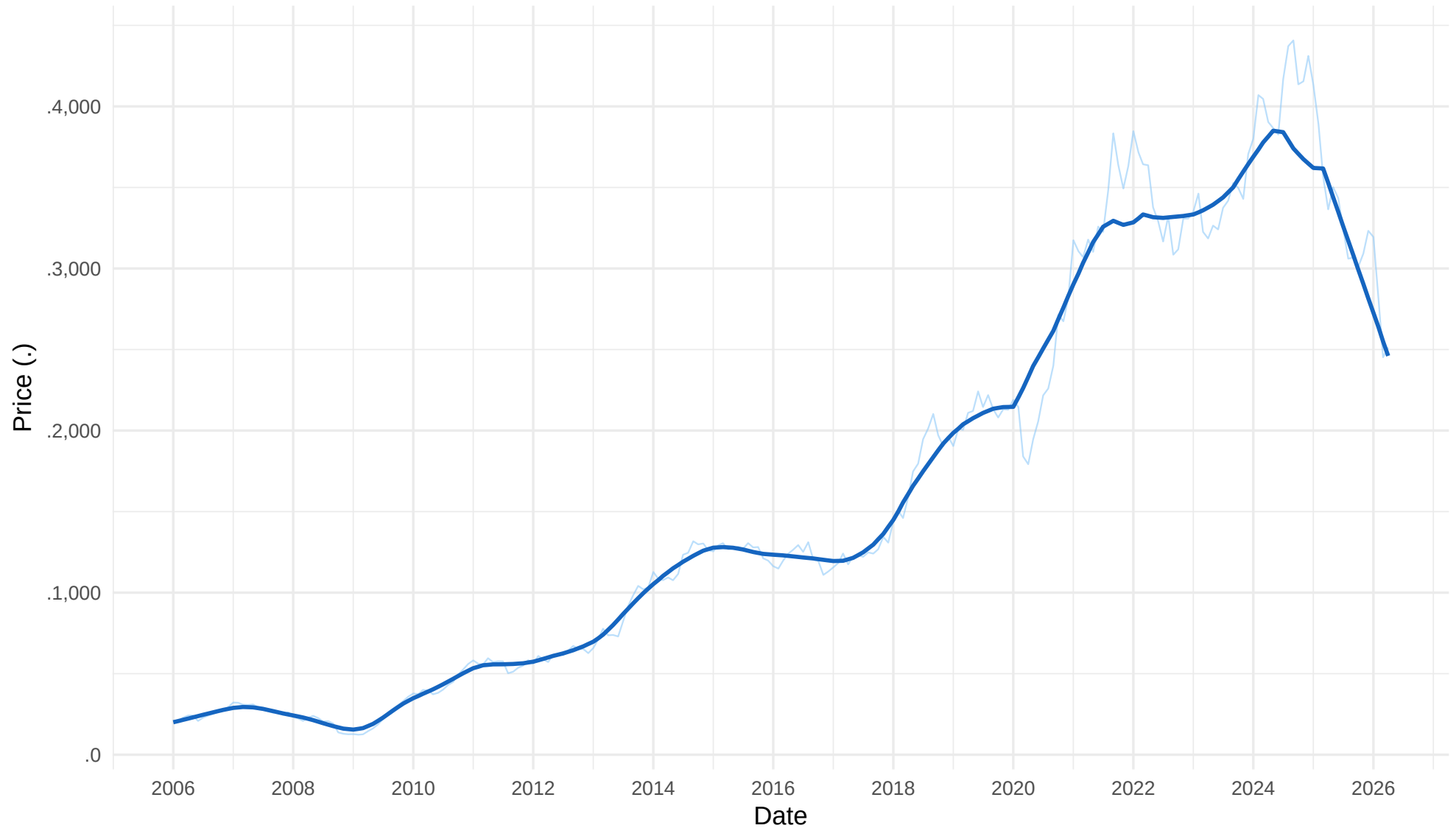
TCS – Seasonal Component

Repeating patterns within each year



TCS – Trend Component

Underlying long-term movement extracted by STL



Light blue = Observed | Dark blue = Trend

TCS Stock – STL Decomposition

